

Electric Circuits Basics

An electric circuit is a closed path that allows electric charge to move through components.

Key facts:

Electric current is the rate of flow of electric charge and is measured in amperes.

Voltage is electric potential difference and is measured in volts.

Resistance opposes current and is measured in ohms.

Ohm's law states that voltage equals current multiplied by resistance: $V = I \times R$.

In a series circuit, the same current flows through each component and resistances add.

In a parallel circuit, components share the same voltage and total resistance decreases as branches are added.

Electrical power is the rate of energy transfer and can be calculated as $P = V \times I$.

Answerable questions:

Q: What is Ohm's law? A: Ohm's law is $V = I \times R$, relating voltage, current, and resistance.

Q: What changes in a parallel circuit? A: Each branch has the same voltage, and adding branches usually lowers total resistance.